HW01 · Ch 21 Coulomb's Law — Worked Solutions

Source: CST homework01 (Chapter 21), mxhuang@scut.edu.cn · study-slides (vetame21.github.io/study-slides)

1 A conductor is distinguished from an insulator with the same number of atoms by the number of:

A) nearly free atoms
B) electrons
C) nearly free electrons ✓ (answer)
D) protons
E) molecules

> **Answer: C**

SOLUTION

What makes a material a conductor is having electrons that can move freely. Both have the same atoms (and so the same electrons and protons); the conductor just has many *nearly free* electrons.

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2 A small object has charge Q. Charge q is removed from it and placed on a second small object. The two objects are placed 1 m apart. For the force that each object exerts on the other to be a maximum, q should be:

A) $2Q$
B) Q
C) $Q/2$ ✓ (answer)
D) $Q/4$
E) 0

> **Answer: C**

SOLUTION

The two charges are $Q - q$ and q, so

$$F = k\frac{(Q-q)\,q}{r^2}.$$

With r fixed, maximize $f(q) = (Q-q)q = Qq - q^2$:

$$\frac{df}{dq} = Q - 2q = 0 \;\Rightarrow\; q = \frac{Q}{2}.$$

Why this method: "maximum" means set the derivative to zero. (It's a downward parabola, so this is a maximum.)

3 **Two small charged objects repel each other with a force F when separated by a distance d. If the charge on each object is reduced to one-fourth of its original value and the distance between them is reduced to $d/2$, the force becomes:**

A) $F/16$
B) $F/8$
C) $F/4$ ✓ (answer)
D) $F/2$
E) F

Answer: C

SOLUTION

$$F' = k\frac{(q_1/4)(q_2/4)}{(d/2)^2} = \frac{1/16}{1/4}\,k\frac{q_1 q_2}{d^2} = \frac{F}{4}.$$

Why this method: use ratios. You never need the actual charges; just track how each factor in Coulomb's law changes.

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4 **A 5.0-C charge is 10 m from a –2.0 C charge. The electrostatic force on the positive charge is:**

A) 9.0×10^8 N toward the negative charge ✓ (answer)
B) 9.0×10^8 N away from the negative charge
C) 9.0×10^9 N toward the negative charge
D) 9.0×10^9 N away from the negative charge
E) none of these

Answer: A

SOLUTION

$F = k\frac{|q_1 q_2|}{r^2} = (9 \times 10^9)\frac{(5)(2)}{10^2} = 9.0 \times 10^8$ N. Opposite signs attract, so it points toward the negative charge.

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5 **A particle with charge Q is on the y axis a distance a from the origin and a particle with charge q is on the x axis a distance d from the origin. The value of d for which the x component of the force on the second particle is the greatest is:**

A) 0
B) a
C) $\sqrt{2}\,a$
D) $a/2$
E) $a/\sqrt{2}$ ✓ (answer)

Answer: E

SOLUTION

The distance is $\sqrt{d^2 + a^2}$ and $\cos\theta = d/\sqrt{d^2 + a^2}$, so

$$F_x = kQq\,\frac{d}{(d^2 + a^2)^{3/2}}.$$

Differentiate:

$$\frac{d}{dd}\left[\frac{d}{(d^2 + a^2)^{3/2}}\right] = \frac{(d^2 + a^2) - 3d^2}{(d^2 + a^2)^{5/2}} = 0 \;\Rightarrow\; a^2 = 2d^2 \;\Rightarrow\; d = \frac{a}{\sqrt{2}}.$$

Why this method: at $d = 0$ the force is all vertical ($F_x = 0$), and far away the force is tiny, so there's a maximum in between; calculus finds it.

6 **A particle with a charge of 5.0×10^{-6} C and a mass of 20 g moves uniformly with a speed of 7 m/s in a circular orbit around a stationary particle with a charge of -5.0×10^{-6} C. The radius of the orbit is:**

A) 0 m

B) 0.23 m ✓ (answer)

C) 0.62 m

D) 1.6 m

E) 4.4 m

Answer: B

SOLUTION

The electric attraction is the centripetal force: $k\frac{Qq}{r^2} = \frac{mv^2}{r} \Rightarrow r = \frac{kQq}{mv^2} = \frac{(9\times10^{9})(25\times10^{-12})}{(0.020)(49)} = 0.23$ m.

7 **Two protons (p_1 and p_2) and an electron (e) lie on a straight line, as shown. The directions of the force of p_1 on e, the force of p_2 on e, and the total force on e, respectively, are:**

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The answer arrows and the figure were images in the original file; official answer D.

Answer: D

SOLUTION

Each proton pulls the electron toward itself (opposite charges attract). The total force points toward the CLOSER proton, because Coulomb's force grows as $1/r^2$.

8 **Particle 1 with charge q_1 and particle 2 with charge q_2 are on the x axis, with particle 1 at $x = a$ and particle 2 at $x = -2a$. For the net force on a third charged particle at the origin to be zero, q_1 and q_2 must be related by $q_2 =$**

A) $2q_1$

B) $4q_1$ ✓ (answer)

C) $-2q_1$

D) $-4q_1$

E) $-q_1/4$

Answer: B

SOLUTION

The two forces must point in opposite directions, so q_1 and q_2 have the SAME sign, and be equal in size: $\frac{q_1}{a^2} = \frac{q_2}{(2a)^2} \Rightarrow q_2 = 4q_1$.

9 **A particle with charge 2 μC is placed at the origin. An identical particle is placed 2 m from the origin on the x axis, and a third identical particle 2 m from the origin on the y axis. The magnitude of the force on the particle at the origin is:**

A) 9.0×10^{-3} N
B) 6.4×10^{-3} N
C) 1.3×10^{-2} N ✓ (answer)
D) 1.8×10^{-2} N
E) 3.6×10^{-2} N

Answer: C

SOLUTION

Each neighbour pushes with $F = k\frac{(2\times10^{-6})^2}{2^2} = 9.0 \times 10^{-3}$ N, one along $-x$ and one along $-y$. At right angles they add to $\sqrt{2}\,F = 1.3 \times 10^{-2}$ N.

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10 **Two particles, each with charge Q, and a third particle with charge q are placed at the vertices of an equilateral triangle as shown. The total force on the particle with charge q is:**

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A) parallel to the left side of the triangle
B) parallel to the right side of the triangle
C) parallel to the bottom side of the triangle
D) perpendicular to the bottom side of the triangle ✓ (answer)
E) perpendicular to the left side of the triangle

Answer: D

SOLUTION

The two equal charges Q sit symmetrically about the vertical line through q. Their sideways components cancel and the vertical ones add, so the total force is perpendicular to the bottom side.

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11 **Charge is distributed uniformly on the surface of a spherical nonconducting shell. A point particle with charge q is inside. The electrical force on the particle q is least when:**

A) it is near the inside surface of the shell
B) it is at the center of the shell
C) it is halfway between the shell center and the inside surface
D) it is near, but not at, the center of the shell
E) all of the above (the force is zero everywhere inside) ✓ (answer)

Answer: E

SOLUTION

Shell theorem: a uniformly charged shell exerts NO net force on a charge anywhere inside it, so the force is zero (and equally 'least') at every point.

12 **Two uncharged metal spheres, L and M, are in contact. A negatively charged rod is brought close to L, but not touching it, as shown. The two spheres are slightly separated and the rod is then withdrawn. As a result:**

> Figure not available: the original homework file was missing when these solutions were made.

A) both spheres are neutral
B) both spheres are positive
C) both spheres are negative
D) L is negative and M is positive
E) L is positive and M is negative ✓ (answer)

Answer: E

SOLUTION

The negative rod pushes free electrons from L over to M. Separating the spheres traps that imbalance: L (near the rod) ends up positive and M negative. This is charging by induction.

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13 **Two identical conducting spheres A and B carry equal charge and exert electrostatic forces of magnitude F on each other. A third identical, uncharged conducting sphere C is first touched to A, then to B, and finally removed. The force between A and B becomes:**

A) $F/2$
B) $F/4$
C) $3F/8$ ✓ (answer)
D) $F/16$
E) 0

Answer: C

SOLUTION

Identical conductors share charge equally. C + A: each gets $q/2$. C + B: $(q/2 + q)/2 = 3q/4$ each. New force $\propto \frac{q}{2} \cdot \frac{3q}{4} = \frac{3}{8}q^2$, so $F' = 3F/8$.

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14 **Which of the following is NOT a possible value for the electric charge on an object?**

A) 8.0×10^{-20} C ✓ (answer)
B) 8.0×10^{-19} C
C) 8.0×10^{-18} C
D) 8.0×10^{-17} C
E) 8.0×10^{-16} C

Answer: A

SOLUTION

Charge comes in whole multiples of $e = 1.6 \times 10^{-19}$ C. 8.0×10^{-20} C $= 0.5e$, which is impossible; the others are $5e$, $50e$, $500e$, $5000e$.

15 **In positron emission tomography, a positron interacts with an atomic electron, emitting gamma rays that are detected by the imager. This gamma ray emission process is called:**

A) pair production

B) annihilation ✓ (answer)

C) absorption

D) alpha decay

E) fission

Answer: B

SOLUTION

$e^- + e^+ \to \gamma + \gamma$ is annihilation (total charge $-e + e = 0$ before and after). Pair production is the reverse, $\gamma \to e^- + e^+$.

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16 **A particle ($m = 50$ g, $q = 5.0$ μC) is released from rest when it is 50 cm from a second particle ($Q = -20.0$ μC). Determine the magnitude of the initial acceleration of the 50 g particle.**

A) 54 m/s^2

B) 90 m/s^2

C) 72 m/s^2 ✓ (answer)

D) 65 m/s^2

E) 36 m/s^2

Answer: C

SOLUTION

$F = k\frac{|qQ|}{r^2} = (9\times10^9)\frac{(5\times10^{-6})(20\times10^{-6})}{0.5^2} = 3.6$ N, then $a = F/m = 3.6/0.050 = 72$ m/s^2.

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17 **Identical point charges Q are placed at each of the four corners of a 3.0 m × 4.0 m rectangle. If $Q = 40$ μC, what is the magnitude of the electrostatic force on any one of the charges?**

A) 3.0 N

B) 2.4 N ✓ (answer)

C) 1.8 N

D) 3.7 N

E) 2.0 N

Answer: B

SOLUTION

With $kQ^2 = (9\times10^9)(40\times10^{-6})^2 = 14.4$ N·m², take the charge at the origin with neighbours at $(3,0)$, $(0,4)$ and $(3,4)$. All forces are repulsive (pointing away), so in magnitude:

$$F_x = 14.4\left(\frac{1}{9} + \frac{1}{25}\cdot\frac{3}{5}\right) = 1.947\text{ N}, \qquad F_y = 14.4\left(\frac{1}{16} + \frac{1}{25}\cdot\frac{4}{5}\right) = 1.361\text{ N},$$

$$|F| = \sqrt{1.947^2 + 1.361^2} = 2.4\text{ N}.$$

Why this method: superposition. Each pair force comes from Coulomb's law, and the diagonal one is split with $\cos\theta = 3/5$, $\sin\theta = 4/5$.

18 **The symbol $\frac{1}{4\pi\varepsilon_0}$ appears in Coulomb's law because we use independently defined units for:**

A) force and distance
B) charge and distance
C) distance and force
D) force, distance and electric charge ✓ (answer)
E) charge

Answer: D

SOLUTION

Newtons, metres and coulombs are each defined on their own, so a constant is needed to make $F = (\text{const})\, q_1 q_2 / r^2$ come out right. That constant is $1/(4\pi\varepsilon_0)$.

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19 **A point charge Q is at the origin and a second charge $2Q$ is on the x axis at $x = -3.0$ m. If $Q = 50$ μC, what is the magnitude of the electrostatic force on a third point charge $-Q$ placed on the y axis at $y = +4.0$ m?**

A) 2.5 N
B) 3.0 N ✓ (answer)
C) 3.7 N
D) 4.4 N
E) 1.8 N

Answer: B

SOLUTION

$kQ^2 = (9\times10^9)(50\times10^{-6})^2 = 22.5$ N·m².

$$F_x = -\frac{2(22.5)}{25}\cdot\frac{3}{5} = -1.08\ \text{N}, \qquad F_y = -\frac{22.5}{16} - \frac{2(22.5)}{25}\cdot\frac{4}{5} = -2.85\ \text{N},$$

$$|F| = \sqrt{1.08^2 + 2.85^2} = 3.0\ \text{N}.$$

Why this method: vector superposition. The 3-4-5 triangle gives the direction cosines for free.

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20 **Two small non-conducting spheres are both charged positively, the combined charge being 40 μC. When the two spheres are 50 cm apart, each is repelled from the other by a force of 2.0 N. Determine the magnitude of the smaller of the two charges.**

A) 1.4 μC ✓ (answer)
B) 1.1 μC
C) 2.0 μC
D) 3.3 μC
E) 17 μC

Answer: A

SOLUTION

$$x(40\times10^{-6} - x) = \frac{(2.0)(0.25)}{9\times10^9} = 5.56\times10^{-11}$$

$$x^2 - (40\times10^{-6})x + 5.56\times10^{-11} = 0 \;\Rightarrow\; x = \frac{40\times10^{-6} \pm \sqrt{1.6\times10^{-9} - 2.22\times10^{-10}}}{2}.$$

The two roots, 1.4 μC and 38.6 μC, are the two charges; the smaller is **1.4 μC**. **Why this method:** two unknowns, two facts (their sum and the force), so substitute and solve the quadratic.

Problem 1. Three charges A, B and C are arranged as shown in the figure. Rank the charges in order of increasing magnitude of the net force F_A, F_B, F_C they experience. Indicate ties where appropriate.

> Figure not available: the original homework file was missing when these solutions were made.

> **Answer:** $C < A < B$

From the official solution (charges of size q, neighbouring spacing d):

$$F_A = k\frac{q^2}{d^2} + k\frac{q^2}{(2d)^2} = \frac{5}{4}\,k\frac{q^2}{d^2}, \quad |F_B| = 2k\frac{q^2}{d^2}, \quad F_C = -k\frac{q^2}{(2d)^2} + k\frac{q^2}{d^2} = \frac{3}{4}\,k\frac{q^2}{d^2}.$$

So $\frac{3}{4} < \frac{5}{4} < 2$: **C < A < B**.

Why this method: Superposition along a line: add the two Coulomb forces with signs (same direction adds, opposite directions subtract).

Problem 2. A charge $q_1 = -6.50$ μC is at the origin, and $q_2 = -2.42$ μC is on the x axis at $x = 1.00$ m. Find the net force on $q_3 = +1.40$ μC located at $x = 0.55$ m. ($k = 9.0 \times 10^9$ N·m²/C²)

> **Answer:** $\vec{F}_3 = -0.12\ \text{N}\,\hat{\imath}$

q_3 is attracted toward both negative charges. Toward q_1 (to the left):

$$F_{31} = k\frac{|q_1 q_3|}{0.55^2} = (9\times10^9)\frac{(6.50\times10^{-6})(1.40\times10^{-6})}{0.3025} = 0.27\ \text{N}\ (-\hat{\imath}).$$

Toward q_2 (to the right, 0.45 m away):

$$F_{32} = (9\times10^9)\frac{(2.42\times10^{-6})(1.40\times10^{-6})}{0.2025} = 0.15\ \text{N}\ (+\hat{\imath}).$$

Net: $-0.27 + 0.15 = -0.12$ N, i.e. **0.12 N toward $-x$**.

Why this method: 1-D superposition: give each force a sign from its direction, then add.

Problem 3. With the same $q_1 = -6.50$ μC at the origin and $q_2 = -2.42$ μC at $x = 1.00$ m, find the position x (between them) where the net force on a test charge q_3 is zero.

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> **Answer:** $x \approx 0.62$ m

Between two like charges the pulls are opposite, so set the magnitudes equal:

$$\frac{|q_1|}{x^2} = \frac{|q_2|}{(1.00-x)^2} \;\Rightarrow\; \frac{1.00-x}{x} = \sqrt{\frac{|q_2|}{|q_1|}} \;\Rightarrow\; x = \frac{1.00\ \text{m}}{1+\sqrt{2.42/6.50}} = 0.62\ \text{m}.$$

It sits closer to the SMALLER charge q_2, as expected.

Why this method: Balance of forces: equal magnitudes, then take square roots (choose the root that lies between the charges).

Problem 4. Three charges are placed as shown: $q_1 = +2.00$ μC and $q_3 = +4.00$ μC; the sign and value of q_2 are unknown. The net force on q_3 is entirely in the $-x$ direction. (a) Find the magnitude of q_2. (b) Find the magnitude of the net force on q_3.

> Figure not available: the original homework file was missing when these solutions were made.

(a)

> **Answer:** $|q_2| = 0.843$ μC (positive)

Only a positive q_2 can make the net force point along $-x$. The y-components must cancel:

$$k\frac{|q_1|q_3}{(0.040)^2}\sin\theta_1 = k\frac{|q_2|q_3}{(0.030)^2}\sin\theta_2 \;\Rightarrow\; q_2 = \frac{9}{16}\,q_1\frac{\sin\theta_1}{\sin\theta_2} = \frac{9}{16}(2.00\ \mu\text{C})\cdot\frac{3}{4} = 0.843\ \mu\text{C}.$$

Why this method: Use the given direction of the net force as an equation: zero y-component.

(b)

> **Answer:** $F = 56.2$ N

Now add the x-components:

$$F = kq_3\left[\frac{q_1}{(0.040)^2}\cdot\frac{4}{5} + \frac{q_2}{(0.030)^2}\cdot\frac{3}{5}\right] = (9\times10^9)(4.00\times10^{-6})\left[1.00\times10^{-3} + 5.62\times10^{-4}\right] = 56.2\ \text{N}.$$

Why this method: With $F_y = 0$, the net force is just F_x.